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溢流导致坝体溃决的数值模拟

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Numerical modelling of dam failure due to flow overtopping

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Abstract A one-dimensional numerical model for dam failure due to flow overtopping is developed. The MacCormack explicit finite difference scheme is used to solve the one-dimensional equations of continuity and momentum for unsteady varied flow over steep bed slopes. In the computation of erosion process, sediment transport equations are considered and the modified Smart formula developed for steep bed slope is selected. The sliding stability of the overtopped dam is checked by modified ordinary method of slices. The model has been successfully calibrated and verified using laboratory experimental data. By comparing with the experimental results, it was found that the model accuracy depends largely on the sediment transport formula and pore water pressure coefficient. The model was found to predict actual breach outflow of the Buffalo Creek Dam reasonably well and closer than other existing numerical models.

Key words dam failure; mathematical modelling; physical models; finite difference scheme; unsteady flow; erosion and sediment transport

Modélisation numérique d'une rupture de barrage due à un débordement

Résumé Nous avons développé un modèle numérique à une dimension de rupture de barrage due à un débordement. Nous avons utilisé le schéma explicite aux différences finies de MacCormack pour résoudre les équations à une dimension de continuité et de conservation des moments pour un écoulement transitoire sur un lit à forte pente. La stabilité du barrage a été testée par une version modifiée de l'habituelle méthode des tranches. Le modèle a été calé et vérifié avec succès sur la base de données expérimentales obtenues au laboratoire. La comparaison avec les données expérimentales a montré que la précision du modèle dépend beaucoup de la formule de transport des sédiments et du coefficient de pression de pore. Il apparaît que le modèle peut prédire de façon réaliste une véritable brèche du barrage de Buffalo Creek plus précisément que d'autres modèles numériques existants.

Mots clefs rupture de barrage; modélisation mathématique; modèles physiques; schéma aux différences finies; écoulement transitoire; érosion et transport des sédiments

INTRODUCTION

Failure of earth dams and embankment dams can be due to various causes such as flow overtopping and piping discharge. Dam failure due to flow overtopping has occurred frequently in the past; however, in-depth knowledge of the mechanism of dam failures and measured data are still lacking. A simulation model of dam failure processes would therefore be useful. Of particular concern in this study is dam failure due to flow overtopping. In the past, most simulation models of sediment transport and river

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溢流导致坝体溃决的数值模拟

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摘要：本文开发了一种用于模拟溢流导致坝体溃决的一维数值模型。采用 MacCormack 显式有限差分格式求解陡坡床面上非恒定变流的连续性方程与动量守恒方程。在冲蚀过程计算中，考虑了泥沙输移方程，并选用了专为陡坡床面开发的修正 Smart 公式。溢流坝体的滑动稳定性采用改进的常规条分法进行验算。该模型已成功利用实验室试验数据完成率定与验证。与试验结果的对比表明，模型精度在很大程度上取决于所采用的泥沙输移公式及孔隙水压力系数。该模型对布法罗溪坝实际溃口出流的预测结果较为合理，且优于其他现有数值模型。

关键词：坝体溃决；数学建模；物理模型；有限差分格式；非恒定流；冲蚀与泥沙输移

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关键词：坝体溃决；数学建模；物理模型；有限差分格式；非恒定流；冲蚀与泥沙输移

引言

土坝与堤坝失事可能由多种原因引起，例如溢流与管涌渗流。历史上曾频繁发生因溢流导致的坝体失事；然而，对其失事机理的深入认识及实测数据仍显不足。因此，建立坝体失事过程的模拟模型十分必要。本研究特别关注由溢流引发的坝体失事问题。过去，多数泥沙输移及河流模拟模型主要面向

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morphology were developed for subcritical flow in rivers and open channels with small bed slopes (Tingsanchali & Supharatid, 1996; Simons & Senturk, 1977). The physical processes of scour and deposition in rivers are different from the cases of dam failures due to flow overtopping. A number of simulation models of dam failures due to flow overtopping have been developed. Fundamentally they can be classified into three types: the first type assumes the breach to have a fixed size equal to the whole dam structure; the second type assumes the breach to have a fixed chosen size; and the third type determines the breach size as a function of time due to erosion, geotechnical aspect of breach stability and hydrology, etc. Meon (1989) further classified the third type models into three groups, namely parametric, physical–parametric and stochastic. The parametric models are such as the Dam Break Flood Forecasting Model of the US National Weather Service (DAMBRK) (Fread, 1979) and the model of the Hydrologic Engineering Center (HEC-1) (Singh & Snorrason, 1984). The physical–parametric models are such as that of Brown & Rogers (1981)—BRDAM, Lou's (1981) model, the model of Ponce & Tsivoglou (1981), the Breach Erosion of Earth-filled Dams and Flood Routing Model (BEED) (Singh & Scarlatos, 1989), NOAA-BREACH (Fread, 1988) and the MIKE-11 Dambreak model (DHI, 1994). An example of the stochastic models is the model of Meon (1989). Of most physical concern are the physical–parametric models. The NOAA-BREACH model considers the breach channel to have sliding failure of side walls within the limit of angle of repose, its bottom slope always parallel to the downstream face of the dam and its bottom width equal to 2–2.5 times the critical depth. Kast & Bieberstein (1997) modified the NOAA-BREACH model incorporating additional details of breach characteristics in the model computation. The MIKE-11 Dambreak model (DHI, 1994) assumes that the breach channel has a horizontal bed and there is no sliding failure of the dam surface. The weakness of these models is that none of them considers the actual mechanisms of dam surface slope failure in the form of slip circle sliding. Moreover, there is no readily available sediment transport formula for calculating erosion of dam surface due to overtopping discharge which has a shallow depth and a Froude number greater than 1 (supercritical flow).

The purpose of this study is to develop a mathematical model which can satisfactorily simulate or predict dam surface erosion and slope sliding failure with time due to flow overtopping. It includes a check on the accuracy of the simulation results by comparing the model results with data from laboratory experiments as well as with observed field data (Singh, 1996) and with the existing numerical models, namely the BREACH model (Fread, 1988) and the MIKE-11 Dambreak model (DHI, 1994). The effects of model parameters on the computed results and the numerical stability are also investigated using sensitivity analysis.

MODEL OF DAM BREACHING DUE TO FLOW OVERTOPPING

The model of dam breaching due to flow overtopping consists of three modules: a module of unsteady flow over steep bed slopes, a dam surface erosion module, and a slope sliding failure module. The model simulation is subdivided into two reaches. Figure 1 shows the model reaches and the flow regime. Model Reach I starts from the far upstream end of the reservoir (Node *a*) to the front edge of the dam crest (Node *c*). Model Reach II starts from the front edge of the dam crest to a station located 2.4 m

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针对小坡度河流与明渠中的缓流，已发展出若干河床形态模型（Tingsanchali & Supharatid, 1996；Simons & Senturk, 1977）。河流中冲刷与淤积的物理过程，与溢流所致坝体溃决的情形存在差异。目前已开发出多种溢流导致坝体溃决的模拟模型。从本质上讲，这些模型可分为三类：第一类假设溃口尺寸固定，等于整个坝体结构；第二类假设溃口尺寸为预先选定的固定值；第三类则根据冲蚀、溃口稳定性的岩土特性、水文等因素，将溃口尺寸作为时间的函数予以确定。Meon（1989）进一步将第三类模型细分为三组，即经验模型、物理—经验模型及随机模型。经验模型包括美国国家气象局（NWS）的坝溃洪水预报模型（DAMBRK）（Fread, 1979）以及水文工程中心（HEC-1）模型（Singh & Snorrason, 1984）。物理—经验模型包括 Brown & Rogers（1981）提出的 BRDAM 模型、Lou（1981）模型、Ponce & Tsivoglou（1981）模型、土石坝溃口冲蚀与洪水演进模型（BEED）（Singh & Scarlatos, 1989）、NOAA-BREACH 模型（Fread, 1988）以及 MIKE-11 坝溃模型（DHI, 1994）。随机模型的典型代表是 Meon（1989）模型。在物理机制方面，尤以物理—经验模型最受关注。NOAA-BREACH 模型假定溃口渠道侧壁发生滑动失稳，其极限为休止角；溃口底坡始终平行于坝体下游面；溃口底宽取临界水深的 2–2.5 倍。Kast & Bieberstein（1997）对 NOAA-BREACH 模型进行了改进，在模型计算中纳入了更多关于溃口特征的细节。MIKE-11 坝溃模型（DHI, 1994）则假定溃口渠道床面水平，且坝面不发生滑动失稳。上述模型的共同缺陷在于：均未考虑以圆弧滑动形式表现的坝面边坡失稳的实际机理；此外，尚无现成可用的泥沙输移公式，可用于计算因溢流泄量（具有浅水深及弗劳德数大于 1 的急流特性）所导致的坝面冲蚀。

rge which has a shallow depth and a Froude number greater than 1 (supercritical flow).

本研究旨在建立一种数学模型，以合理模拟或预测因溢流而导致的坝面冲蚀及边坡滑动失稳随时间的发展过程。该模型通过将模拟结果与实验室试验数据、实测现场数据（Singh, 1996）以及现有数值模型（即 BREACH 模型（Fread, 1988）和 MIKE-11 坝溃模型（DHI, 1994））进行对比，对模拟结果的精度进行校验。此外，还采用敏感性分析方法，考察模型参数对计算结果及数值稳定性的影响。

溢流导致坝体溃决的模型

溢流导致坝体溃决的模型由三个模块组成：陡坡床面非恒定流模块、坝面冲蚀模块以及边坡滑动失稳模块。模型模拟划分为两个河段。图 1 示出模型河段及其水流流态。模型河段 I 起始于水库远上游端（节点 a），终止于坝顶前沿（节点 c）；模型河段 II 起始于坝顶前沿，终止于距坝趾下游 2.4 米处的断面

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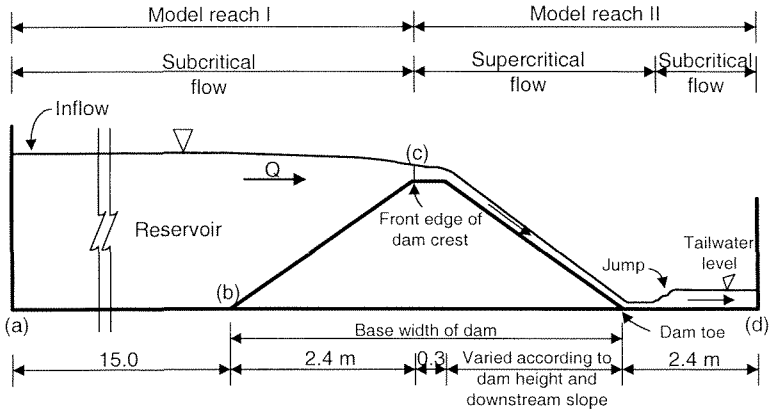


Fig. 1 Model reaches and flow regime.

downstream of the dam toe (Node *d*). For Model Reach I, the module of slope sliding failure, is not considered. For Model Reach II, all three modules are considered. The theory and numerical computation of each module are explained below:

Module for unsteady flow over steep bed slopes

The one-dimensional system of equations referred to as steep-slope shallow water equations can be applied for the case of flow over a dam. In vector form, the governing equations are:

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} = S \tag{1}$$

in which the vectors U, F, and S are given by the following equations:

$$U = \begin{bmatrix} h \\ uh \end{bmatrix} \tag{2}$$

$$F = \begin{bmatrix} uh \\ u^2 h + \frac{1}{2}(gh^2 \cos\theta) \end{bmatrix} \tag{3}$$

$$S = \begin{bmatrix} 0 \\ gh(\sin\theta - S_f) \end{bmatrix} \tag{4}$$

where h is the depth of flow, u is the flow velocity, g is the acceleration due to gravity, θ is the angle of the channel bed, S_f is the slope of energy grade line, x is the longitudinal distance, and t is time.

In this study, the MacCormack explicit finite difference scheme (MacCormack, 1971) is used which is second-order accurate in space and time. Its main advantage is its ability to simultaneously calculate gradually varied and rapidly varied flow. The finite difference forms of the governing equations are as follows:

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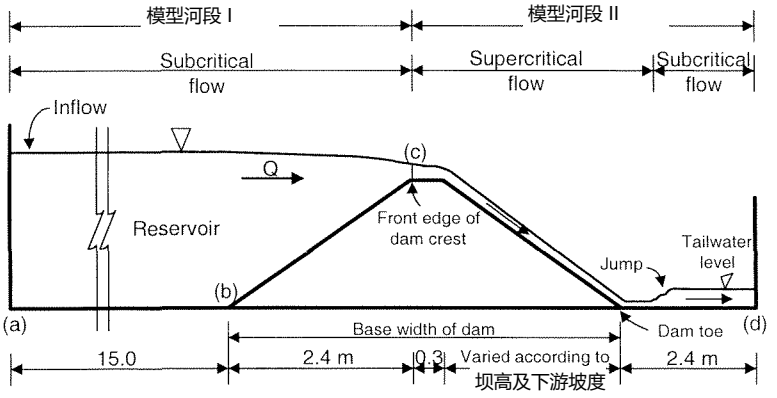


图 1 模型河段及水流流态

坝趾下游（节点 d）。对于模型河段 I，不考虑边坡滑动失稳模块；对于模型河段 II，则考虑全部三个模块。各模块的理论基础与数值计算方法如下所述：

陡坡床面非恒定流模块

所称陡坡浅水方程的一维方程组可应用于坝顶溢流情形。其控制方程以向量形式表示为：

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} = S \tag{1}$$

其中向量 U、F 和 S 由下列方程给出：

$$U = \begin{bmatrix} h \\ u \end{bmatrix} \tag{2}$$

$$F = \begin{bmatrix} u_i \\ u\ h\ H\text{—}(gh\cos\theta) \end{bmatrix} \tag{3}$$

$$S = \begin{bmatrix} 0 \\ gh\sin\theta - Q\,S_f \end{bmatrix} \tag{4}$$

其中h为水深，u为流速，g为重力加速度，θ为渠床倾角，S_f为能量坡度，x为纵向距离，t为时间。

本研究采用MacCormack显式有限差分格式（MacCormack, 1971），该格式在空间与时间上均为二阶精度。其主要优势在于可同步计算渐变流与急变流。控制方程的有限差分形式如下：

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Predictor step: $$U_i^p = U_i^k - \frac{\Delta t}{\Delta x}\left(F_{i+1}^k - F_i^k\right) + \Delta t S_i^k \tag{5}$$

Corrector step: $$U_i^{k+1} = \frac{1}{2}\left(U_i^k + U_i^p\right) - \frac{\Delta t}{2\Delta x}\left(F_i^p - F_{i-1}^p\right) + \frac{1}{2}\Delta t S_i^p \tag{6}$$

where the superscript p refers to the predictor (intermediate) step, k is the time step and i is the computational mesh point.

Numerical computation At the start of the computation, the necessary initial conditions for the unsteady free surface flow equations are that all the flow velocities and water surface elevations at every grid point along the reach must be known. Corresponding to an initial overtopping discharge, the measured water surface elevations from the experiment in this study are specified at all grid points along the considered channel. In Model Reach I, the flow in the reservoir is subcritical. Therefore, the upstream boundary condition is the constant reservoir inflow. The downstream boundary condition is the observed depth hydrograph at the front edge of the dam crest, which is moving upstream with time due to headcut erosion process. For Model Reach II, the flow over the dam crest and along the downstream dam surface is supercritical. Just downstream of the dam toe, a hydraulic jump occurs and subcritical flow exists. Therefore, two boundary conditions must be given at the upstream end of Model Reach II, namely the computed discharge hydrograph from Model Reach I and the observed depth hydrograph at the front edge of the dam crest. For the subcritical flow at the end of Model Reach II, the downstream boundary is specified by assuming a zero discharge gradient for the last two nodes. The downstream flow depth is obtained using the forward finite difference scheme.

Numerical stability is directly associated with explicit numerical computation. The MacCormack explicit finite difference scheme is stable if the Courant-Friedrichs-Lewy (CFL) condition is satisfied. The condition is given by:

$$\Delta t \le C_n \frac{\Delta x}{\mathrm{Max}\left(|u| + \sqrt{gh}\right)} \tag{7}$$

where C_n is the flow Courant number and its value must be less than 1 for numerical stability.

Module for dam surface erosion

The erosion of the dam crest and the surface of the dam can be calculated by the sediment continuity equation and the sediment transport formula. The sediment continuity equation can be written as follows:

$$\frac{\partial Q_S}{\partial x} + (1-p)B\frac{\partial z}{\partial t} = 0 \tag{8}$$

where Q_s is the solid volume of total sediment transport rate, p is the porosity of bed material, B is the wetted perimeter of movable bed, z is the bed elevation, x is the distance, and t is the time.

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预估步：$U_i^p = U_i^k - (\Delta t/\Delta x)(F_i^k - F_i^k) + \Delta t S_i$ (5)

校正步：$U_i^{k1} = ½(U_i^p + U_i^k) - (\Delta t$ (6) 2Δv ?

其中上标p表示预估（中间）步，k为时间步，i为计算网格点。

数值计算：计算起始时，非恒定自由水面流方程所需的初始条件为：沿整个计算河段各网格点处的流速与水面高程均需已知。对应于初始漫顶流量，本研究试验实测的水面高程被指定为所考虑渠道各网格点的初始值。在模型河段I中，水库内水流为缓流，因此上游边界条件取为恒定入库流量；下游边界条件取为坝顶前沿观测到的水深过程线，该过程线因溯源冲刷而随时间向上游移动。在模型河段II中，坝顶及下游坝面水流为急流；坝趾下游紧邻处发生水跃，此后为缓流。因此，在模型河段II上游端需同时给定两个边界条件，即由模型河段I计算所得的流量过程线，以及坝顶前沿观测所得的水深过程线；在模型河段II末端的缓流区，下游边界通过假定最后两个节点处流量梯度为零来设定，下游水深则采用前向有限差分格式求得。

数值稳定性与显式数值计算直接相关。MacCormack显式有限差分格式在满足Courant-Friedrichs-Lewy（CFL）条件时稳定。该条件表述如下：

$$\Delta t \le C_n \frac{\Delta x}{\mathrm{Max}\left(|u| + \sqrt{gh}\right)} \tag{7}$$

其中C为水流柯朗数，为保证数值稳定性，其值须小于1。

坝面侵蚀模块

坝顶及坝面侵蚀可通过泥沙连续性方程与泥沙输移公式计算。泥沙连续性方程可表述如下：

$$\partial z/\partial t + (1-p)B\,\partial Qs/\partial x = 0 \tag{8}$$

其中Qs为总泥沙输移率的固体体积，p为床沙孔隙率，B为可动床湿周，z为床面高程，x为距离，t为时间。

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The continuity equation of sediment transport can be written in an explicit finite difference form using the modified Lax scheme (De Vries, 1987) as follows:

$$\frac{\partial z}{\partial t} = \frac{1}{\Delta t}\left\{ z_i^{j+1} - \left[(1-\alpha) z_i^j + \alpha \frac{z_{i+1}^j + z_{i-1}^j}{2} \right] \right\} \qquad 0 \le \alpha \le 1 \tag{9}$$

$$\frac{\partial Q_S}{\partial x} = \frac{Q_{S\,i+1}^{\ j} - Q_{S\,i-1}^{\ j}}{2\Delta x} \tag{10}$$

where Δt is the time step, Δx is the space step, and α is the numerical weighting coefficient which controls the stability and accuracy of the numerical results. The upper index j indicates the time; the lower index i is the position along the x-axis.

For a wide rectangular channel with solid walls, the wetted perimeter B can be replaced by the channel width. The finite difference equation of the sediment continuity equation for the unit width of dam surface can be expressed as:

$$z_i^{j+1} = \left[(1-\alpha) z_i^j + \alpha \frac{z_{i+1}^j + z_{i-1}^j}{2} \right] - \frac{\Delta t}{2(1-p)\Delta x} \left(q_{S\,i+1}^{\ j} - q_{S\,i-1}^{\ j} \right) \tag{11}$$

where q_s is the volumetric sediment discharge per unit channel width. The stability condition for the numerical model is given by De Vries (1987) as follows:

$$\mu^2 < \alpha < 1 \tag{12}$$

where $\mu = c_3 \Delta t / \Delta x$ is the sediment transport Courant number and c_3 is the bed celerity in a frictionless system, which is expressed by (De Vries, 1987):

$$c_3 = \frac{u}{h\left(1 - F^2\right)} \frac{\mathrm{d}q_s}{\mathrm{d}u} \tag{13}$$

where u is the flow velocity, h is the depth of flow, q_s is the sediment transport rate per unit width, and F is the Froude number.

For the sediment transport equation, it can be written in a general form as:

$$q_s = \mathrm{f}\,(u, h, d, n, \ldots, \text{etc.}) \tag{14}$$

where u is the flow velocity, h is the depth of flow, d is the grain size diameter, and n is Manning's roughness coefficient.

For the computation of dam surface erosion, the initial conditions are the given initial bed or dam surface elevations. The upstream boundary condition is the sediment load–discharge relationship according to the specified sediment transport formula.

Module of slope sliding failure

The safety factor of a potential failure surface is defined as a ratio between the resisting force and the driving force, both applied along the failure surface. Due to the interaction between strengths and forces, the failure mass is divided into a series of vertical slices and the equilibrium condition of each of these slices is considered as shown in Fig. 2. These forces include the resultants X and E of shear and normal

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采用修正Lax格式（De Vries, 1987），泥沙输移连续性方程可写为如下显式有限差分形式：

$$\frac{\partial z}{\partial t} = \frac{1}{\Delta t}\left\{ z_i^{j+1} - \left[(1-a)z_i + a : 0 < a < 1 \right. \right. \tag{9}$$

$$\frac{dQs}{dx} = \frac{QsU - QsU}{2\Delta A -} \tag{10}$$

其中Δt为时间步长，Δx为空间步长，a为数值加权系数，用于控制数值结果的稳定性与精度；上标j表示时间层，下标i表示沿x轴的位置。

对于壁面固定的宽矩形渠道，湿周B可用渠道宽度代替。单位宽度坝面泥沙连续性方程的有限差分形式可表示为：

$$\left[z_i^j + \alpha \frac{z_{i+1}^j + z_{i-1}^j}{2} \right] - \frac{\Delta t}{2(1-p)\Delta x} \left(q_{S\,i+1}^{\,j} - (Js) \right) \tag{11}$$

其中qs为单位渠道宽度的体积输沙率。数值模型的稳定性条件由De Vries（1987）给出如下：

$$u^2 < a < 1 \tag{i²}$$

其中u = cΔt/Δx为泥沙输移柯朗数，c为无摩擦系统中床面传播速度，其表达式见De Vries（1987）：

$$u \rightarrow qs \qquad h(1-F^2)\, du / C \tag{13}$$

其中u为流速，h为水深，qs为单位宽度输沙率，F为弗劳德数。

泥沙输移方程可写为如下一般形式：qs = f(u, h, d, n, ...)

(14)

其中u为流速，h为水深，d为颗粒粒径，n为曼宁糙率系数。

在计算坝面冲刷时，初始条件为给定的初始河床或坝面高程；上游边界条件依据指定的泥沙输移公式确定，即输沙率与流量之间的关系。

边坡滑动失稳模块

潜在滑动面的安全系数定义为沿该滑动面作用的抗滑力与下滑力之比。由于强度与力之间的相互作用，滑动体被划分为一系列垂直切片，并考虑每一垂直切片的静力平衡条件，如图2所示。这些力包括剪应力和正应力的合力X与E

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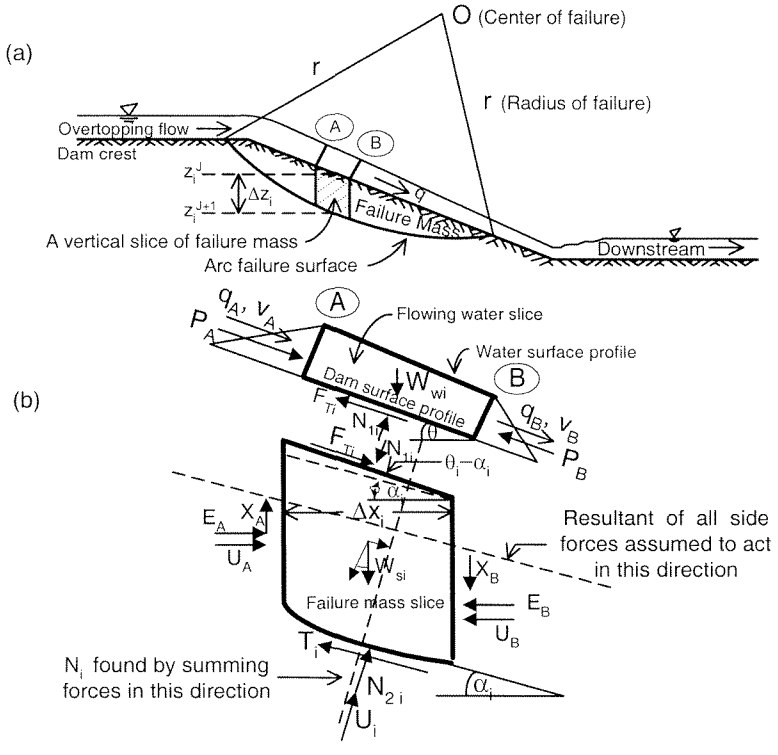


Fig. 2 (a) Arc failure surface with flow overtopping and (b) forces considered in ordinary method of slices.

effective forces along the side of the slide, as well as resultants T_i and N_{2i} of the shear and normal effective forces acting along the failure arc. Also acting on this slice are the resultants U_A and U_B of the pore water pressures against the sides of the slice and U_i against the failure arc. In addition, due to overtopping water forces, F_{Ti} and N_{1i} are the tractive force acting along the surface of contact between flowing water and the dam bed, and the normal force acting on the dam bed. The safety factor is defined in terms of moments about the centre of failure arc:

$$FS = \frac{M_R}{M_D} = \frac{\text{Moment of shear strength along failure arc}}{\text{Moment of driving forces}} \tag{15}$$

Based on the ordinary method of slices (Fellenius, 1936), it is assumed that the forces acting upon the sides of any slice have zero resultant in the direction normal to the failure arc for that slice as shown in Fig. 3 ($E_A = E_B$, $X_A = X_B$, $U_A = U_B$). With this assumption,

$$\overline{N}_{2i} = W_{Si}\cos\alpha_i + \overline{N}_{1i}\cos(\theta_i - \alpha_i) + F_{Ti}\sin(\theta_i - \alpha_i) - U_i \tag{16}$$

where $\overline{N}_{2i}$ is the resultant of the normal effective force acting along the failure arc, U_i is the boundary water force, W_{Si} is the weight of a failure slice, N_{1i} is the normal force acting on the dam bed = $W_{wi}\cos\theta_i$, W_{wi} is the weight of overtopping water, θ_i is the

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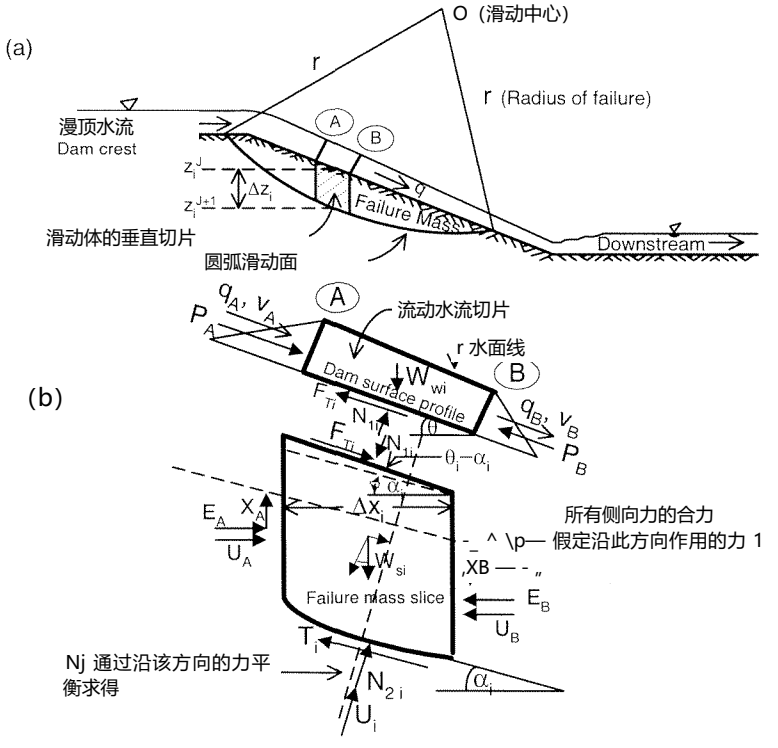


图2 （a）漫顶水流作用下的圆弧溃决面；（b）普通条分法中考虑的力系。

滑动条块侧向的有效力，以及沿溃决圆弧作用的抗剪力与有效正向力的合力7/和iNi。此外，作用于该条块的还有条块两侧孔隙水压力的合力U,\和UD，以及溃决圆弧上的Uj。另外，由于漫顶水流的作用，FT,和N],分别为水流与坝基接触面上的拖曳力，以及作用于坝基的正向力。安全系数以对溃决圆弧中心的力矩定义：

$$FS = \text{MR 溃决圆弧上抗剪强度力矩；M,, 驱动力矩} \qquad (15)$$

依据普通条分法（Fellenius, 1936），假定任一条块两侧所受力在该条块溃决圆弧法向的合力为零，如图3所示（EA = EB，Xt\ = XR，U,\ = UB）。在此假设下，

$$\text{Ni VFs,cos}\alpha, + \text{NM cos } (\theta,. - \alpha ,) + F\, r\, ; \sin(-\alpha,. U; \qquad (16)$$

其中A72,为沿溃决圆弧作用的有效正向力合力，U,为边界水压力，W$,为溃决条块重量，A^,-为作用于坝基的正向力 = WV,-cosθ,-，WKI为漫顶水重，θ,为

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angle of bed slope, α_i is the angle of failure slice, i is the number of a failure slice, and F_{Ti} is the friction force or tractive force acting along the surface of contact between water and the dam surface. By assuming hydrostatic pressure distribution, F_{Ti} can be given as:

$$F_{T_i} = \frac{1}{2}\gamma\left(h_A^2 - h_B^2\right) + \rho\left[(qv)_A - (qv)_B\right] + W_{w_i}\sin\theta_i \tag{17}$$

where γ is the specific weight of water, h is the flowing water depth, ρ is the density of water, q is the discharge per unit width, v is the flow velocity, A and B represent section indices.

The resisting moment may be written as:

$$M_R = r\sum_{i=1}^{i=n}\left(\bar{c}\Delta x/\cos\alpha_i + \tan\bar{\phi}\,\overline{N}_{2i}\right) \tag{18}$$

where r is the radius of failure arc, n is the number of slices, $\bar{c}$ is the soil cohesion based on effective stresses, Δx is the width of a slice, α_i is the angle of failure slice, and $\bar{\phi}$ is the friction angle based on effective stresses.

Also, the resultant of the shear force acting along the failure arc may be written as

$$T_i = W_{Si}\sin\alpha_i - \overline{N}_{1i}\sin(\theta_i - \alpha_i) + F_{Ti}\cos(\theta_i - \alpha_i) \tag{19}$$

Similarly, the driving moment may be written as:

$$M_D = r\sum_{i=1}^{i=n} T_i = r\sum_{i=1}^{i=n}\left[W_{Si}\sin\alpha_i - \overline{N}_{1i}\sin(\theta_i - \alpha_i) + F_{Ti}\cos(\theta_i - \alpha_i)\right] \tag{20}$$

In this study, the dam is made of noncohesive material in which $\bar{c} = 0$. Substituting equations (18) and (20) into equation (15), by assuming that $\bar{\phi}$ is constant along the failure arc, the safety factor becomes:

$$FS = \frac{\tan\bar{\phi}\sum_{i=1}^{i=n}\overline{N}_{2i}}{\sum_{i=1}^{i=n}\left[W_{Si}\sin\alpha_i - \overline{N}_{1i}\sin(\theta_i - \alpha_i) + F_{Ti}\cos(\theta_i - \alpha_i)\right]} \tag{21}$$

It is assumed that at the failure instant, the flow overtopping penetrates downward through the dam surface to the arc of failure with the depth approximately equal to the water depth above it. With this assumption, the pore water pressure in the dam body is:

$$U = Cu\,\Sigma\gamma_i\,\Delta h_i \tag{22}$$

where C_u is the empirical pore water pressure coefficient, γ_i is the unit weight of soil at slice i, and Δh_i is the vertical distance from the free surface of the overtopping water to the elevation of arc circle of failure at the slice i.

The coordinates of the centre of failure circle are assumed by trial and error until the arc of failure is found. When the safety factor of a potential failure surface is less than one, the failure slice masses are assumed transported by the overtopping flow.

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床面坡度角，α,为溃决条块倾角，i为溃决条块序号，Fn为沿水流与坝体表面接触面作用的摩阻力或拖曳力。假定静水压力分布，则Fn可表示为：

$$FT = \gamma \times h \times lil - h\ ) + \rho[\{qv\}A - \{qv\}B\ ] + Wlr \sin \theta, \tag{17}$$

其中γ为水的容重，h为流动水深，ρ为水的密度，q为单宽流量，v为流速，A和B代表断面编号。

抗滑力矩可表示为：

$$M\ R = \sum_{i=1}^{i=n} r\hat{}(c\Delta x/\cos\alpha, + \tan\varphi \times Ni,) \tag{18}$$

其中r为溃决圆弧半径，n为条块数量，c为基于有效应力的土体黏聚力，Δx为单个条块宽度，α,为溃决条块倾角，φ为基于有效应力的内摩擦角。

此外，沿溃决圆弧作用的抗剪力合力可表示为：7; = WSi sin α, - JVI, sin(θ, - α,.) + FTi cos(θ, - α ,) (19)

同理，驱动力矩可表示为：

$$MD = \sum_{i=1}^{i=n} r\hat{}T, = \sum_{i=1}^{i=n} rjr[ws,.sina. - iVi,sin(\theta, - \alpha ,) + FTi \cos(\theta, -\alpha,.)] \tag{20}$$

本研究中，坝体由无黏性材料构成，即c = 0。将式(18)和式(20)代入式(15)，并假定溃决圆弧上φ沿程恒定，则安全系数变为：

$$FS = \frac{\alpha \sum_{i=1}^{i=n} \tan\hat{}JT\hat{I}Vi;}{\sum_{i=1}^{i=n} p , , . . \sin,. - iVi, - \sin(\theta,. - \alpha,.) + F\ n\cos(\theta, - \alpha,.)J} \tag{21}$$

假定在溃决发生瞬间，漫顶水流向下穿透坝体表面直至溃决圆弧，其渗透深度近似等于其上方水深。据此假设，坝体内孔隙水压力为：

$$U = Cu \times y; \times Ah \tag{22}$$

其中C„为经验性孔隙水压力系数，y,为第i个条块处土体的单位重，Ah,为漫顶水流自由水面至该条块处溃决圆弧高程的垂直距离。

溃决圆弧中心坐标通过试错法假定，直至确定溃决圆弧。当某一潜在溃决面的安全系数小于1时，假定该溃决条块的质量由漫顶水流搬运。

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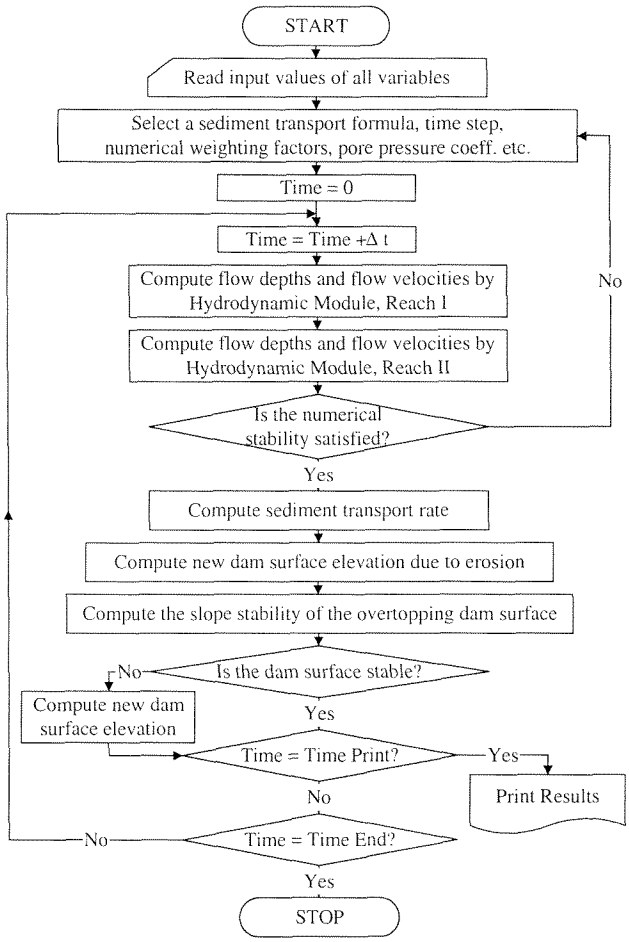


Fig. 3 Flow chart of the numerical computation.

The new surface elevation of the arc circle of failure at slice i is $z_i^{j+1} = z_i^j - \Delta z_i$; where z_i^{j+1} is the dam surface elevation of the failure slice i at time $j+1$, z_i^j is the dam surface elevation of slice i at time j, and Δz_i is the change in the dam surface elevation of the failure slice i. Figure 3 shows the flowchart of the numerical computation in the model of dam breaching due to flow overtopping.

EXPERIMENTS

Experiments were carried out in a rectangular flume 35 m long, 1.0 m deep, and 1.0 m wide. The height and crest width of the dam were fixed at 0.80 m and 0.30 m. Figure 4 shows the experimental setup. The upstream slope was fixed at 1V:3H, while the downstream (D/S) slopes were varied. Two sizes of the medium sand were selected: i.e. Sand I with D_{30} = 0.52 mm, D_{50} = 0.86 mm, mean grain size D_m = 1.13 mm, and

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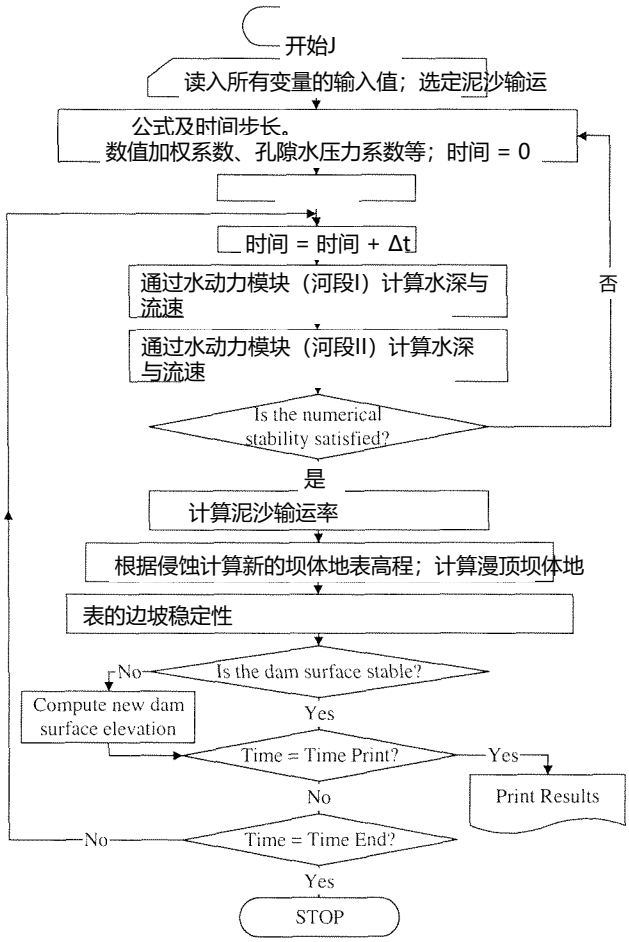


图3 数值计算流程图

第i个滑动条块在j+1时刻的溃决弧线新地表高程为z/+1 = z- - Az;；其中z/rl为第i个溃决条块在j+1时刻的坝体地表高程，zj为第i个条块在j时刻的坝体地表高程，Az,为第i个溃决条块坝体地表高程的变化量。图3展示了漫顶水流导致坝体溃决模型中数值计算的流程图。

实验

实验在一条长35 m、深1.0 m、宽1.0 m的矩形水槽中进行。坝体高度和坝顶宽度分别固定为0.80 m和0.30 m。图4展示了实验装置。上游坡比固定为1V:3H，下游（DIS）坡比则变化设置。选取了两种粒径的中等砂：即砂I，其D30 = 0.52 mm，D50 = 0.86 mm，平均粒径D„, = 1.13 mm，以及

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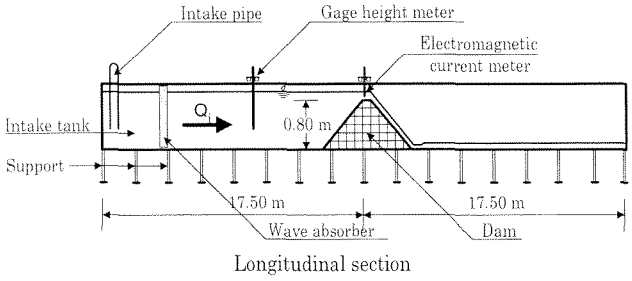


Fig. 4 Experimental setup.

Table 1 Summary of the parameters used in the experiments.

Experiments	Run no.	Material	Downstream slope	Reservoir inflow ($l\ s^{-1}$)	Depth* (cm)
Fixed bed	A-1	Sand I	1V:3H	5.35	-
	A-2	Sand I	1V:3H	26.25	-
	A-3	Sand II	1V:3H	5.40	-
	A-4	Sand II	1V:3H	18.50	-
Movable bed: dam surface erosion	B-1	Sand I	1V:5H	1.05	3
	B-2	Sand I	1V:5H	1.58	5
	B-3	Sand II	1V:5H	5.10	5
Movable bed: dam surface erosion and sliding failure	C-1	Sand I	1V:3H	1.42	3
	C-2	Sand I	1V:2.5H	1.23	3
	C-3	Sand II	1V:2.5H	1.60	3
	C-4	Sand II	1V:3H	2.78	3

* Based on initial upstream water level above dam crest.

D_{90} = 3.80 mm; and Sand II with D_{30} = 0.26 mm, D_{50} = 0.36 mm, mean grain size D_m = 0.30 mm, and D_{90} = 0.58 mm. Table 1 summarizes the characteristics of dams and hydraulic conditions used in all experiments. The details of each experiment are explained as follows:

Experiments under fixed bed condition (Case A)

In the experiments, a fixed bed dam made of plywood was constructed in the middle reach of the flume. A medium sand was glued over the plywood plate. Six wave gauges were used to measure the water level along the dam surface. A two-dimensional electromagnetic current meter was installed at the middle of the dam crest to measure the overtopping flow velocity. A 90° V-notched weir was provided at the end of flume to measure the discharge. Four experimental runs were made.

Movable bed experiments: dam erosion (Case B)

An embankment dam of medium sand was built in the middle reach of the flume. For dam compaction control, the material was water sprayed, and compacted in layers each

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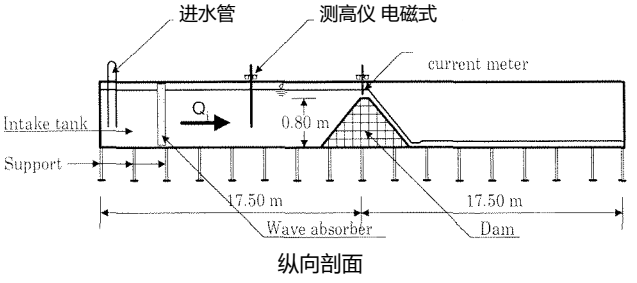


图4 实验装置。

表1 试验所用参数汇总。试验

	试验编号	材料	下游坡比 1V:3H	水库入流量 (L/s)	水深 * (cm)
固定床	A-1	砂料I		5.35	-
	A-2	砂料I	1V:3H	26.25	-
	A-3	砂料II	1V:3H	5.40	-
	A-4	砂料II	1V:3H	18.50	-
可动床：坝面侵蚀	B-1	砂料I	1V:5H	1.05	3
	B-2	砂料I	1V:5H	1.58	5
	B-3	砂料II	1V:5H	5.10	5
可动床：坝面侵蚀与滑动失事	C-1	砂料I	1V:3H	1.42	3
	C-2	砂料I	1V:2.5H	1.23	3
	C-3	砂料II	1V:2.5H	1.60	3
	C-4	砂料II	1V:3H	2.78	3

* 基于初始上游水位高于坝顶的高度。

砂料I：D90 = 3.80 mm；砂料II：D30 = 0.26 mm，D50 = 0.36 mm，平均粒径D50 = 0.30 mm，D90 = 0.58 mm。表1汇总了全部试验所用坝体特性及水力条件。各次试验详情说明如下：

固定床条件下的试验（A工况）

试验中，在水槽中游段修建一座胶合板固定床坝体，中等粒径砂料粘贴于胶合板表面。共布置六支波高仪沿坝面测量水位；在坝顶中部安装二维电磁流速仪测定溢流流速；水槽末端设置90°三角形薄壁堰测量流量。共进行四次试验。

可动床试验：坝体侵蚀（B工况）

在水槽中游段修筑一座中等粒径砂料坝体。为控制坝体压实度，施工中对填料喷水并分层压实，每层

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of 5 cm height. A constant inflow at the upstream end of the flume was supplied. The upstream water level of the dam was increased until breaching developed. Three experimental runs were made. In order to determine the dam surface evolution and the water surface level along the downstream slope of the embankment surface, square grids of 5 cm × 5 cm were drawn on the flume glass walls and a smaller grid of 1 cm × 1 cm was drawn in the region along the dam surface. To have a uniform overflowing across the flume width, a vertical plate was held at the dam crest across the flume width until the upstream water level was higher than the dam crest 3 cm or 5 cm. Then the vertical plate was suddenly lifted up to allow overflow to start. The dam surface evolution was both videotaped and photographed. The flow velocities at the centre of the dam just upstream of the breach were measured continuously by a two-dimensional electromagnetic current meter. Also, the depth hydrographs at the upstream reservoir were continuously recorded with time.

Experiments involving dam surface erosion and sliding failure (Case C)

The dam was built under the same conditions as Case B, except for the downstream slope of the dam, which was steeper than in Case B. The procedures of the experiment were the same as for Case B. Four experimental runs were made.

RESULTS AND DISCUSSION

The model simulation results are divided into three cases, namely: fixed bed condition, dam erosion, and combination of dam erosion and sliding failure. In the mathematical model simulation, the model was calibrated and then verified by using experimental data. The model was further verified with observed field data and other existing numerical models. The results were analysed to study the model performance and the model sensitivity due to variation of parameters.

For numerical stability, the flow Courant number C_n in equation (7) is larger than the sediment Courant number μ in equation (12). Therefore, the computation time step is controlled by the flow Courant number, C_n. For Model Reach I, the space step Δx of 0.15 m was used from Node *a* to Node *b* and Δx was reduced to 0.05 m from Node *b* to Node *c* as shown in Fig. 1. For Model Reach II, the space step Δx was fixed at 0.05 m. For stability in the hydrodynamic computation, the allowable time step determined according to the CFL condition in equation (7) varied from 0.001 to 0.005 s.

Model results under fixed bed condition

Model calibration The hydrodynamic model was calibrated by using experimental data under a fixed bed condition. The data of Runs A-2 and A-4 were used for dam surface material I and II, respectively. The Manning roughness coefficient determined by trial and error is equal to 0.018 and 0.013 for sand material I and II, respectively. In the steady uniform flow, comparisons between the measured and computed water surface profiles showed a good agreement with a normalized root mean square error of about 3% and a correlation coefficient of 0.99.

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高度为5 cm。在水槽上游端施加恒定入流。持续抬升坝体上游水位，直至发生溃口。共进行三次试验。为获取坝面演变过程及坝体下游坡面上的水面线，水槽玻璃侧壁上绘制5 cm × 5 cm方格网，坝面区域则绘制1 cm × 1 cm密格网。为确保沿水槽全宽均匀溢流，在上游水位高于坝顶3 cm或5 cm前，于坝顶横跨水槽设置竖直挡板；随后迅速提起挡板启动溢流。坝面演变过程同时采用录像与摄影记录。溃口上游坝体中心处流速由二维电磁流速仪连续测定；上游水库水位过程线亦随时间连续记录。

涉及坝面侵蚀与滑动失事的试验（C工况）

该坝体建造条件与B工况相同，仅下游坡比更陡。试验流程与B工况一致，共进行四次试验。

成果与讨论

模型模拟成果分为三类情形：固定床条件、坝体侵蚀及坝体侵蚀与滑动失事耦合。在数学模型模拟中，模型先经率定再行验证，并进一步结合现场观测资料及其他既有数值模型开展验证。分析结果旨在评估模型性能及参数变化引起的模型敏感性。

为保障数值稳定性，式（7）中水流Courant数Cf大于式（12）中泥沙Courant数Cs。因此，计算时间步长由水流Courant数Cf控制。模型河段I中，节点a至b的空间步长Δx取0.15 m，b至c段则减小至0.05 m（见图1）。模型河段II中，空间步长Δx固定为0.05 m。为保障水动力计算稳定性，依据式（7）CFL条件确定的允许时间步长在0.001–0.005 s之间变化。

固定床条件下的模型成果

模型率定 水动力模型在固定床条件下，利用试验A-2与A-4的数据进行率定，分别对应坝面材料I与II。经试错法确定的曼宁糙率系数分别为0.018（砂料I）与0.013（砂料II）。在恒定均匀流条件下，实测与计算水面线对比显示良好吻合，归一化均方根误差约3%，相关系数达0.99。

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Model verification The hydrodynamic model was verified using experimental data of Runs A-1 and A-3. All parameters of numerical model were kept the same as in the model calibration except the reservoir inflow. It was found that the model simulates the flows over steep bed slopes with a normalized root mean square error of about 5% and a correlation coefficient of 0.99.

Sensitivity analysis To study the effect of Manning's n, the data of Run A-2 were used. When Manning $n = 0.025$ was used, instead of 0.018, it was found that the water depth increased and flow velocity decreased. When Manning $n = 0.010$ was used, it was found that the water depth decreased and flow velocity increased and the computed point of starting the hydraulic jump was farther than the observed.

Model results for dam erosion

Model calibration The dam erosion model was calibrated by using experimental data of Runs B-1 and B-3 of the movable bed experiments. In this study, five sediment transport formulas, namely Meyer-Peter & Mueller (1948), Smart (1984), Bagnold-Visser (Visser, 1988), Rickenmann (1991), and Takahashi (1991) were selected for calculating the erosion process along the dam surface due to overtopping. Compared to the experimental results, it was found that these formulas are not directly applicable in this study. The Meyer-Peter & Mueller formula, developed for rather coarse sand, shows less erosion owing to lack of parameters relating to the slope effect in the equation. The Bagnold-Visser formula also shows lower estimate of erosion. This formula is more suitable for computing total sediment load of fine sand, which is mainly suspended load transport, while the transport of sediment observed in the experiment is mainly bed load. The Smart formula, developed for bed load transport over a range of steep slope from 0.04% to 20% and Froude number less than 3.0, overestimates dam erosion. The Rickenmann formula, developed for bed load transport over steep slopes (0.1–20%) and Froude number less than 3.0, also overestimates the erosion process. For both the Smart and Rickenmann formulas, dam failure occurs faster than the observed situation in the experiments. The Takahashi formula, developed for debris flow over steep slopes, yields very overestimated results and leads to unstable results. It should be noted that none of the sediment transport formulas studied have been tested to such extreme conditions of erosion process. For the calibration of the dam surface erosion module, the numerical weighting factor α in equation (9) was found to be equal to 0.010 and 0.003 for Model Reaches I and II.

In this study, the Smart (1984) formula was selected because it gave the best results among the others and the effect of slope term is included, which can be modified for further improvement. Smart proposed a bed load transport equation in steep slope flume tests based on the Meyer-Peter & Mueller formula (1948). The Smart formula can be expressed as follows:

$$\phi_B = 1.28\left(D_{90} / D_{30}\right)^{0.2} \left(R^{1/6} / n\right)\theta^x \left(\theta - \theta_{cr}\right) S^y \tag{23}$$

where ϕ_B is the dimensionless sediment transport rate, $\phi_B = q_B / \sqrt{g\left(s-1\right) D_m^3}$, θ is the dimensionless shear stress at the bed, $\theta = h\,S / \left(\Delta\, D_m\right)$, q_B is the volumetric sediment

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模型验证 水动力模型采用A-1与A-3次试验数据进行验证。除水库入流量外，数值模型其余参数均与模型率定阶段一致。结果表明，模型对陡坡水流的模拟效果良好，归一化均方根误差约为5%，相关系数达0.99。

敏感性分析 为研究曼宁n值的影响，采用A-2次试验数据。当n取0.025（而非0.018）时，计算得水深增大、流速减小；当n取0.010时，则水深减小、流速增大，且计算所得水跃起始位置较实测位置更远。

坝体侵蚀模型成果

模型率定 坝体侵蚀模型采用可动床试验B-1与B-3的实测数据进行率定。本研究共选取五种泥沙输移公式——Meyer-Peter & Mueller（1948）、Smart（1984）、Bagnold-Visser（Visser，1988）、Rickenmann（1991）及Takahashi（1991）——用于计算溢流作用下坝面侵蚀过程。与试验结果对比表明，上述公式均无法直接适用。Meyer-Peter & Mueller公式针对较粗砂开发，因公式中缺乏反映坡度效应的参数，导致估算侵蚀量偏低。Bagnold-Visser公式同样低估侵蚀量，其更适用于细砂全沙输移（以悬移质为主），而本试验观测到的泥沙输移则以推移质为主。Smart公式适用于坡度0.04%–20%、弗劳德数小于3.0的推移质输移，但高估了坝体侵蚀量。Rickenmann公式适用于坡度0.1%–20%、弗劳德数小于3.0的推移质输移，亦高估侵蚀过程。对Smart与Rickenmann两种公式而言，坝体失事发生时间均早于试验实测情况。Takahashi公式专为陡坡泥石流开发，结果严重高估且计算不稳定。需指出的是，前述所有泥沙输移公式均未在如此极端的侵蚀工况下经过验证。在坝面侵蚀模块率定中，式（9）中数值加权因子α经确定，在模型河段I与II中分别取0.010与0.003。

本研究选用Smart（1984）公式，因其在所对比公式中表现最优，且已包含坡度项，便于后续改进。Smart基于Meyer-Peter & Mueller公式（1948），通过陡坡水槽试验提出了推移质输移公式。该公式表达如下：

$$\Phi B = 1.28\,(D90 / D30)^{2}\,(R / D50)^{0.5}\,(\theta - \theta c)^{1.5} \tag{23}$$

式中ΦB为无量纲泥沙输移率，$\Phi g = qB / [\sqrt{(g\,(s - 1)\,D50)}]$，θ为床面无量纲剪切应力，$\theta = h\,S / (\Delta\,D50)$，qB为体积泥沙输移率

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discharge per unit channel width (solid), s is the ratio of sediment density to water density, D_m is the mean grain diameter, h is the flow depth, S is the channel bed slope, D_{90} and D_{30} are the grain sizes at which, respectively, 90 and 30% by weight of material is finer, R is the hydraulic radius, n is the Manning flow resistance factor, x and y are the empirical power coefficients which equal 0.5 and 0.6 respectively for channel slopes from 0.04 to 20%, and θ_{cr} is the critical value of θ at initiation of motion which can be expressed as (Stevens *et al.*, 1976):

$$\theta_{cr} = \kappa\theta_{o_{cr}} \text{ where } \kappa = \cos\beta\left(1 - \tan\beta / \tan\phi\right) \tag{24}$$

In equation (24), $\theta_{o_{cr}}$ is the critical Shields' parameter for horizontal bed, κ is the bed slope coefficient for $\theta_{o_{cr}}$, β is the angle of local bed slope, and ϕ is the angle of repose of the submerged bed material.

The dam erosion model calibration consists of two steps: (a) adjustment of the values of the empirical power coefficients of x and y in the original Smart sediment transport, equation (23); and (b) adjustment of the numerical weighting factor α in equation (11). The adjustment is done by trial and error until the best agreement between the computed and measured reservoir water levels, overtopping discharges, and dam bed profiles are obtained. It was found that within the experimental range in this study, the values of x and y in equation (23) are 0.5 and 0.8 respectively for channel slopes from 0 to 18% and are 0.5 and 1.5 for channel slopes steeper than 18%. As shown in Fig. 5, the agreement between the measured and computed reservoir water levels, overtopping discharges, water surface and dam surface profiles were found to be good. For the dam surface profiles, the normalized root mean square error of 4% and a correlation coefficient of 0.99 were found.

Model verification In verifying the numerical model, the data from experiment al Run B-2 are used. In this experiment, the initial reservoir water level was kept to 5 cm above the dam crest by a vertical plate before allowing overflow to occur. The reservoir inflow was 1.58 l s^{-1}. All parameters of the numerical model were kept the same as in the model calibration. From the comparison between the measured and

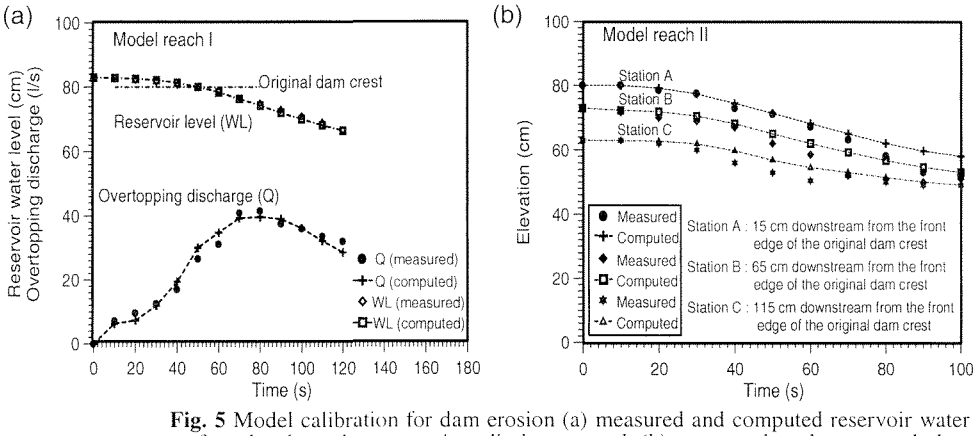


Fig. 5 Model calibration for dam erosion (a) measured and computed reservoir water surface levels and overtopping discharges and (b) measured and computed dam surface levels.

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q为单位渠道宽度流量（实线），s为泥沙密度与水密度之比，D50为平均粒径，h为水流深度，S为渠床坡度，D90和D30分别为累计质量百分数达90%和30%所对应的粒径，R为水力半径，n为曼宁糙率系数，x和y为经验幂系数，在渠床坡度0.04%–20%范围内分别取0.5和0.6；θc为泥沙起动时θ的临界值，可按Stevens等（1976）提出的公式表达：

$$\theta c = K\theta c，其中K = \cos\beta\,(1 - \tan\beta / \tan\varphi) \tag{24}$$

式（24）中，θc为水平床面起动临界Shields参数，K为θc的床面坡度修正系数，β为局部床面坡角，φ为浸没状态下坝体材料的休止角。

坝体侵蚀模型率定包含两个步骤：（a）调整原始Smart泥沙输移公式（式23）中经验幂系数A和y的取值；（b）调整式（11）中的数值加权因子α。该调整通过试错法完成，直至计算所得水库水位、溢流量及坝面高程剖面与实测结果达到最佳吻合。研究发现，在本试验所涉范围内，式（23）中x和y的取值分别为0.5和0.8（适用于渠床坡度0%–18%），而当渠床坡度大于18%时，则分别为0.5和1.5。如图5所示，实测与计算所得水库水位、溢流量、水面线及坝面高程剖面之间吻合良好。对于坝面高程剖面，归一化均方根误差为4%，相关系数达0.99。

模型验证 在数值模型验证过程中，采用试验B-2的实测数据。该试验中，在发生溢流前，通过竖直挡板将水库初始水位维持在坝顶以上5 cm处。水库入流量为1.58 L/s。数值模型的所有参数均与模型率定阶段保持一致。实测值与

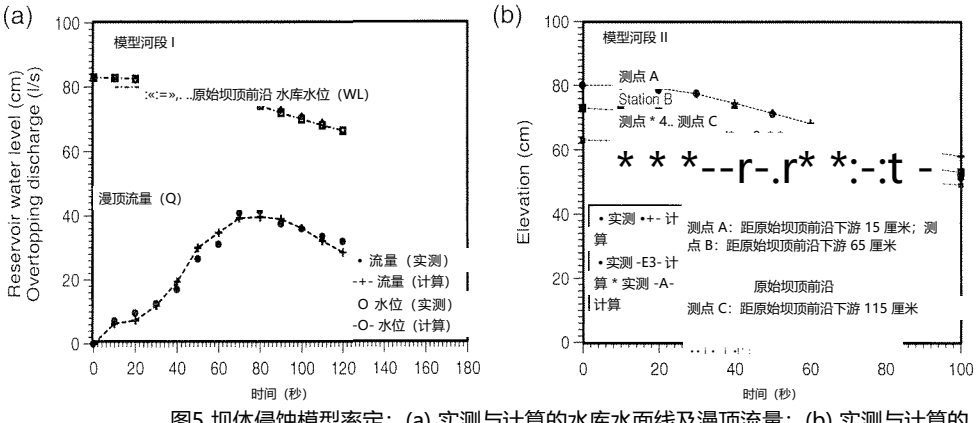


图5 坝体侵蚀模型率定：(a) 实测与计算的水库水面线及漫顶流量；(b) 实测与计算的坝面高程。

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computed overtopping discharge, water surface and dam surface profiles downstream of the crest, the agreement was found to be satisfactory. For the dam surface profiles, the normalized root mean square error of about 9% and a correlation coefficient of 0.98 were found.

Sensitivity analysis The experimental data of Run B-1 were used in the study. In order to analyse the effect of the Manning roughness coefficient on the computed results, the values of the roughness coefficient n = 0.010, 0.018, and 0.025 were used in the computation. When Manning n = 0.010 was used (originally n = 0.018), it was found that the water depth slightly decreased and the flow velocity increased. Hence, more erosion occurred, especially at the downstream region of dam crest in which the flow velocity is high due to steep slopes. The opposite condition occurs when Manning n is increased, e.g. n = 0.025. However, the effect of Manning n on the computed change in bed level was found to be small.

Considering Model Reach II, the effect of the distance-weighting factor in the numerical scheme α in equation (11) was determined first by putting α = 0, 0.003, 0.005, and 0.010. Figure 6(a) shows the effect of α on the computed dam surface profiles. It was found that α has an influence on the results especially on the numerical stability and accuracy of the results. When α = 0, the model was unstable in some cases. When $\alpha > 0$, the model was stable and erosion increased with α. When α was increased, e.g. α = 0.010, the computed bed changes also increased.

The effect of the mean grain size on the computed dam surface profiles was studied by considering sand material with D_{30} = 0.26 mm, D_m = 0.30 mm, and D_{90} = 0.58 mm (originally D_{30} = 0.52 mm, D_m = 1.13 mm, D_{90} = 3.8 mm). It was found that when a smaller grain diameter was used, the computed dam surface profiles changed and the erosion process was faster than that of the originally computed values.

Figure 6(b) shows the effect of bed slope on the critical bed shear stress. The critical Shields' parameter for steep slopes is considered as described in equation (24). At the initial stage of flow overtopping, i.e. time = 30 s, it was found that the critical bed shear stress has an effect on the sediment transport rate along the downstream dam surface. Near the dam toe, where bed slope is rather flat, the effect of bed slope and the

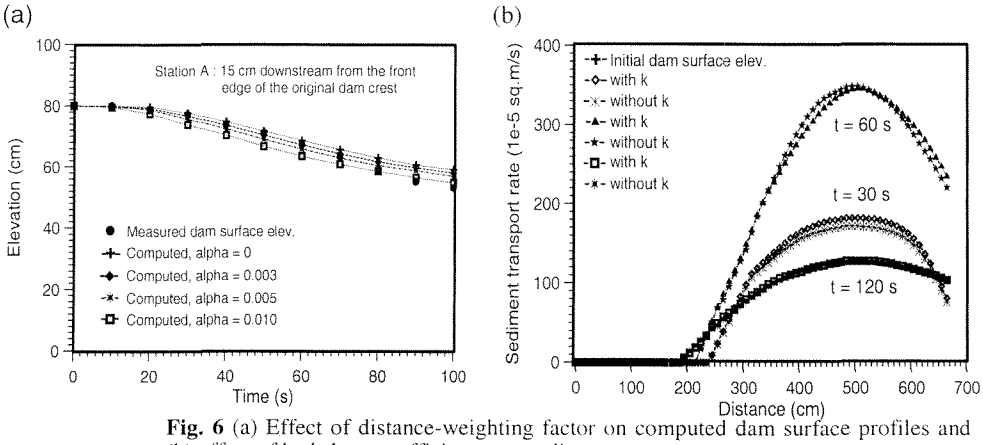


Fig. 6 (a) Effect of distance-weighting factor on computed dam surface profiles and (b) effect of bed slope coefficient κ on sediment transport rate.

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计算所得漫顶流量、水面线及坝顶下游坝面剖面与实测结果吻合良好。对于坝面剖面，归一化均方根误差约为9%，相关系数达0.98。

敏感性分析 本研究采用试验工况 B-I 的实测数据。为分析曼宁糙率系数对计算结果的影响，在计算中分别采用了糙率系数 n = 0.010、0.018 和 0.025。当采用曼宁 n = 0.010（原始取值为 n = 0.018）时，发现水深略有减小、流速增大，因此侵蚀加剧，尤其在坝顶下游区域——该处因坡度较陡导致流速较高。当曼宁 n 增大（例如 n = 0.025）时，则出现相反情况。然而，曼宁 n 对计算所得床面高程变化的影响较小。

针对模型河段 II，首先通过设定 α = 0、0.003、0.005 和 0.010，确定数值格式中公式(11)的距离加权因子 α 的影响。图6(a)显示了 α 对计算坝面剖面的影响。结果表明，α 对计算结果具有一定影响，尤其体现在数值稳定性和计算精度方面。当 α = 0 时，模型在某些情况下不稳定；当 α > 0 时，模型趋于稳定，且侵蚀量随 α 增大而增大；当 α 增大（例如 α = 0.010）时，计算所得床面变化量亦随之增大。

通过采用 D = 0.26 毫米、D = 0.30 毫米、D = 0.58 毫米（原始值为 D = 0.52 毫米、D = 1.13 毫米、D = 3.8 毫米）的砂料，研究了平均粒径对计算坝面剖面的影响。结果表明，当采用更小粒径时，计算所得坝面剖面发生变化，且侵蚀过程比原始计算结果更快。

图6(b) 显示了床面坡度对临界床面剪切应力的影响。陡坡条件下的临界Shields参数按公式(24)描述。在漫顶水流初始阶段（即时间为30秒时），发现临界床面剪切应力对坝下游面的泥沙输移率具有影响。在坝趾附近，床面坡度较平缓，床面坡度及

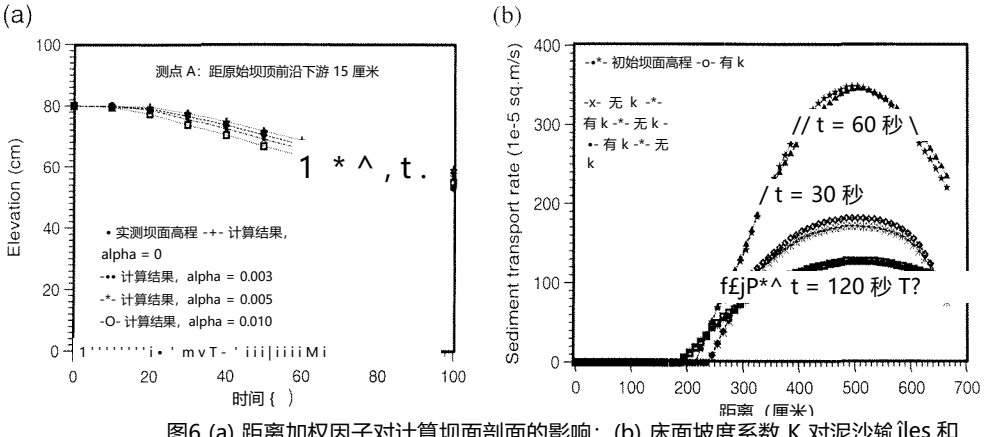


图6 (a) 距离加权因子对计算坝面剖面的影响；(b) 床面坡度系数 K 对泥沙输 îles 和移率的影响。

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critical bed shear stress is small. When the effect of bed slope is considered in the critical bed shear stress computation, the sediment transport rate is higher than in the case without considering the bed slope. At time = 60 s after overtopping occurred, the effect of bed slope in the computation of the critical bed shear stress was reduced. At 120 s, the effect of the bed slope on the critical bed shear stress was insignificant.

Model results for dam erosion with sliding failure

Model calibration During an overtopping event, two types of failure modes were observed along the downstream dam slope, i.e. erosion mode and the slope sliding stability mode. The data of Experiments C-2 and C-4 were used in calibrating the model. In the computation of slope stability of the dam, the parameter C_u (equation (22)), which represents the pore water pressure at the base of the failure slice was adjusted. By trial and error, the best agreement between the computed and measured reservoir water levels, overtopping discharges, and dam surface profiles was obtained when C_u was 0.80. The other parameters such as the numerical weighting factor α were as same as those found in the dam erosion model calibration. For the case C-2, Fig. 7(a) shows the comparison between the measured and computed reservoir water

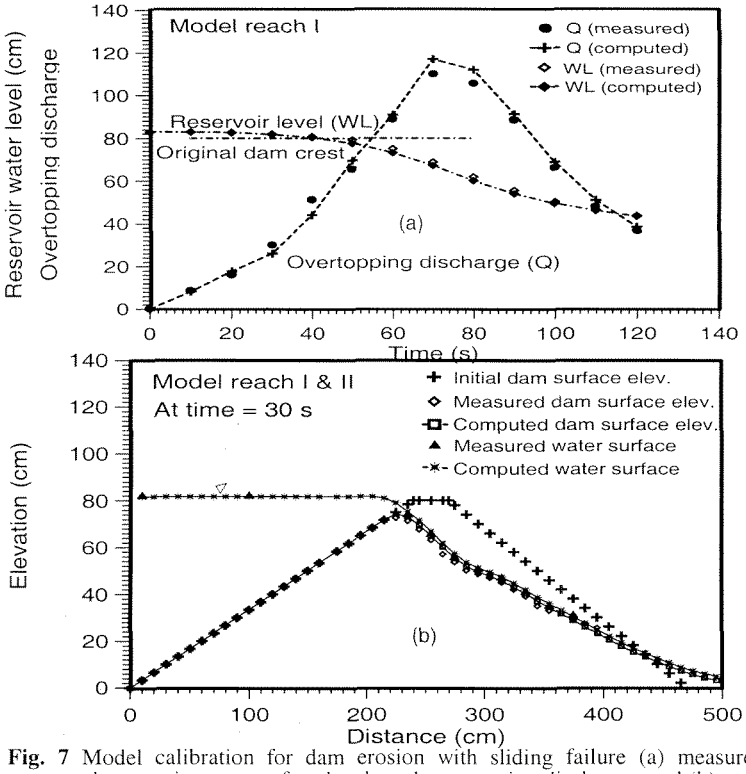


Fig. 7 Model calibration for dam erosion with sliding failure (a) measured and computed reservoir water surface levels and overtopping discharges and (b) measured and computed water surface levels and dam surface profiles.

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临界床面剪切应力较小。当在临界床面剪切应力计算中考虑床面坡度影响时，泥沙输移率高于未考虑床面坡度的情形。漫顶发生后t = 60 s时，床面坡度对临界床面剪切应力计算的影响减弱；t = 120 s时，床面坡度对临界床面剪切应力的影响已可忽略。

含滑动失稳的坝体侵蚀模型计算结果

模型率定 在漫顶过程中，下游坝坡上观察到两类失稳模式：侵蚀模式与边坡滑动稳定性模式。采用C-2与C-4试验数据进行模型率定。在坝体边坡稳定性计算中，调整参数Cu（公式(22)），该参数代表失稳土条底部的孔隙水压力。经反复试算，当Cu取0.80时，计算所得水库水位、漫顶流量及坝面轮廓与实测值吻合最佳。其余参数（如数值加权因子α）均沿用坝体侵蚀模型率定时所确定的取值。对于C-2工况，图7(a)展示了实测与计算的水库水

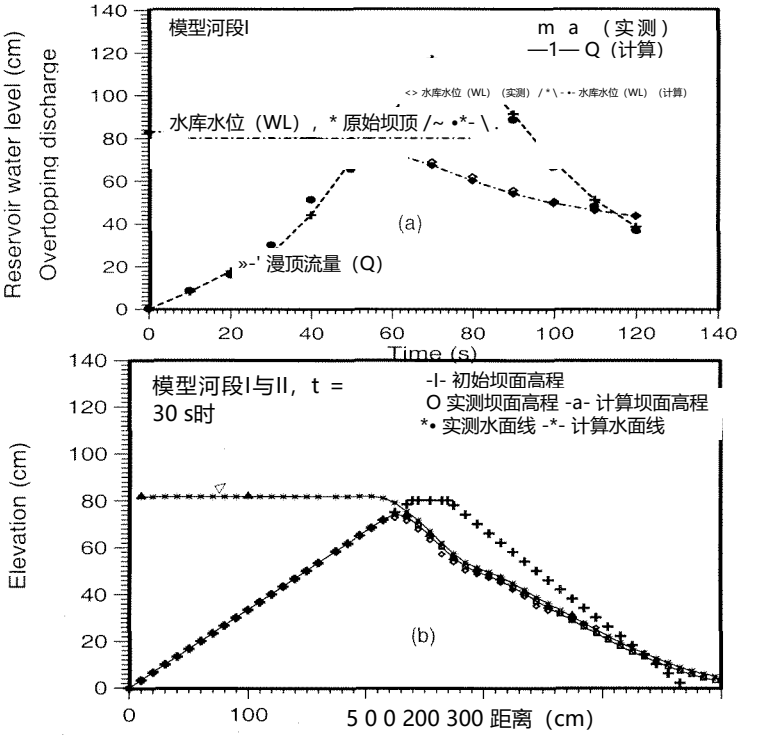


图7 含滑动失稳的坝体侵蚀模型率定：(a) 实测与计算的水库水位及漫顶流量；(b) 实测与计算的水面线及坝面轮廓。

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surface levels and the overtopping discharges. Figure 7(b) shows the comparison between the measured and computed water surface levels and dam surface profiles. For the dam surface profiles, the normalized root mean square error is about 9% and the correlation coefficient is about 0.98.

Model verification In the model verification, data from Experiments C-1 and C-3 were used. All parameters of the numerical model were kept the same as in the model calibration. From the comparison between the measured and computed overtopping discharges, water surface and dam surface profiles downstream of the crest, the agreement was found to be satisfactory as shown in Fig. 8 (for Run C-1). For the dam surface profiles, the normalized root mean square error was 10% and the correlation coefficient was about 0.98.

Sensitivity analysis The model simulation was done for the case C-2 without considering slip circle sliding failure. Compared to the case considering slip circle sliding, the model results show significant increases in errors. The normalized root mean square error of the dam surface profile is increased from 9% to 32% and the correlation coefficient was reduced from 0.98 to 0.90. For the computed peak breach outflow, the normalized root mean square error was increased from 7% to 32% and the correlation coefficient was reduced from 0.99 to 0.87. In the computation of dam stability, pore water pressure is a cause of driving moments that initiate dam surface sliding failure. To study the effect of the empirical pore water pressure coefficient, experimental data of Run C-2 were used. By putting C_u = 0, 0.70, 0.80, and 0.90, it was found that C_u had a significant influence on the accuracy of the results as indicated in Fig. 9. When sliding stability was not considered in the computation, i.e. for C_u = 0, it was found that the computed dam surface erosion was significantly less than the measured data. When C_u = 0.7, the dam surface failure due to sliding was increased, when C_u = 0.8, the computed dam surface profile agreed closely with the measured

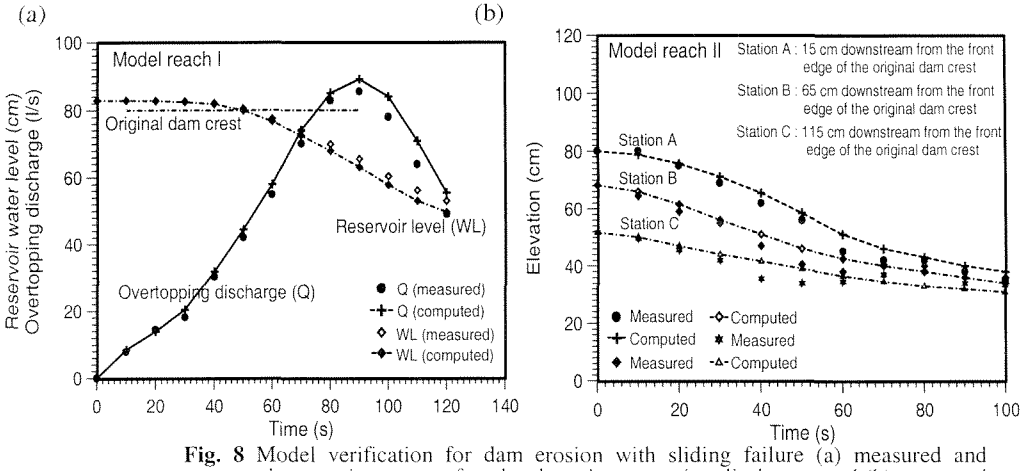


Fig. 8 Model verification for dam erosion with sliding failure (a) measured and computed reservoir water surface levels and overtopping discharges and (b) measured and computed dam surface levels.

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水面线及漫顶流量。图7(b)展示了实测与计算的水面线及坝面轮廓对比。坝面轮廓的归一化均方根误差约为9%，相关系数约为0.98。

模型验证 在模型验证中，采用C-1和C-3试验数据。数值模型所有参数均保持与模型率定阶段一致。通过对比实测与计算的漫顶流量、坝顶下游水面线及坝面轮廓，结果显示二者吻合良好，如图8（C-1工况）所示。坝面轮廓的归一化均方根误差为10%，相关系数约为0.98。

敏感性分析 模型模拟针对C-2工况开展，未考虑圆弧滑动失稳。与考虑圆弧滑动失稳的工况相比，模型结果误差显著增大：坝面轮廓的归一化均方根误差由9%增至32%，相关系数由0.98降至0.90；计算所得溃口峰值流量的归一化均方根误差由7%增至32%，相关系数由0.99降至0.87。在坝体稳定性计算中，孔隙水压力是引发坝面滑动失稳的驱动力矩成因。为研究经验性孔隙水压力系数的影响，采用C-2试验实测数据进行分析。分别取Cu = 0、0.70、0.80和0.90进行计算，结果表明Cu对计算精度具有显著影响，如图9所示。当计算中不考虑滑动稳定性（即Cu = 0）时，计算所得坝面侵蚀量明显小于实测值；当Cu = 0.7时，滑动引起的坝面失稳加剧；当Cu = 0.8时，计算所得坝面轮廓与实测结果高度吻合

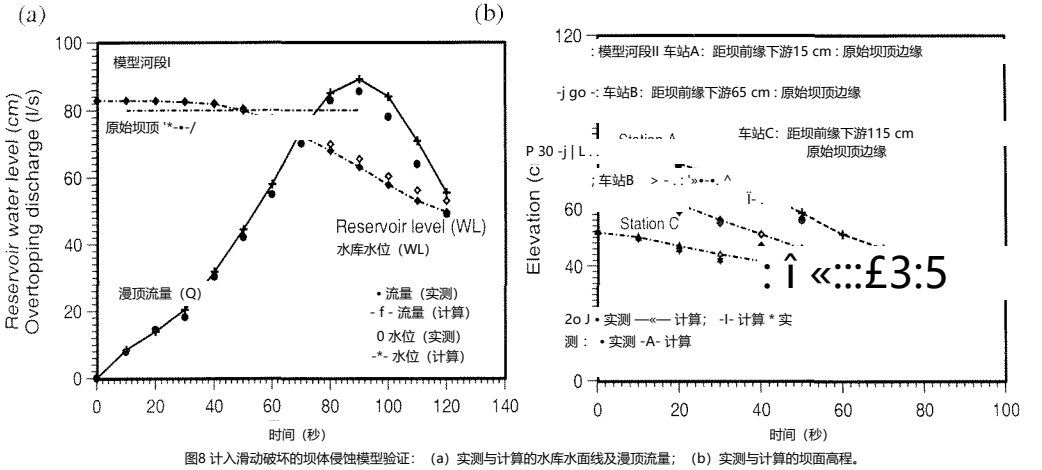


图8 计入滑动破坏的坝体侵蚀模型验证：（a）实测与计算的水库水面线及漫顶流量；（b）实测与计算的坝面高程。

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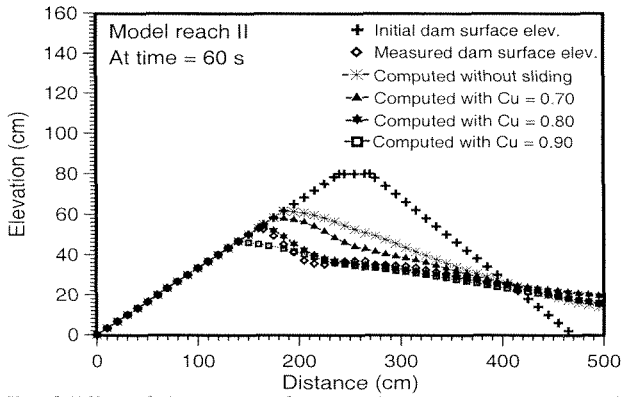


Fig. 9 Effect of slope sliding failure and pore water pressure coefficient on computed dam surface profiles.

data, and when $C_u = 0.9$, the computed dam surface failure overestimated the measured data.

Application of model to actual dam breach

The model was applied to compute dam breach outflow due to overtopping failure of the Buffalo Creek Dam, North Carolina, USA according to the measured data mentioned by Singh (1996). The dam has a height of 32 m, length of 107 m, crest width of 4 m, upstream slope of 1V:3H and downstream slope of 1V:2H. The observed maximum breach outflow was 1420 $m^3 s^{-1}$, which occurred 30 min after overtopping started. By reducing the value of exponent y of the slope S in the modified Smart formula in equation (23) from $y = 0.8$ to 0.6 when $S < 0.18$, the present model computes the dam breach outflow hydrograph with a peak equal to 1.056 of the observed peak and the time to peak of 28 min. The NOAA-BREACH model (Fread, 1988) was also applied in this case. Based on the modified Meyer-Peter & Mueller (1948) formula, the BREACH model computed the breach outflow hydrograph with a much higher peak equal to 2.8 times the observed peak and a shorter time to peak equal to 0.866 times the observed value. Based on the Engelund & Hansen (1967) total load formula, the MIKE-11 model computed the breach outflow hydrograph with a peak equal to 0.167 times the observed peak and a time to peak equal to 1.167 times the observed value.

CONCLUSIONS

In this study, the erosion process of dam due to flow overtopping is considered. The dam material is medium sand. The overtopping section is rectangular and the effect of tail water is negligible. When flow overtopping occurs, erosion starts at the front edge of the dam crest and critical flow occurs at a section just downstream of the front edge of the dam crest. Upstream of this section, the flow is subcritical and there is virtually no erosion. Downstream of this critical flow section, which is the downstream dam surface, the flow becomes supercritical and severe erosion occurs.

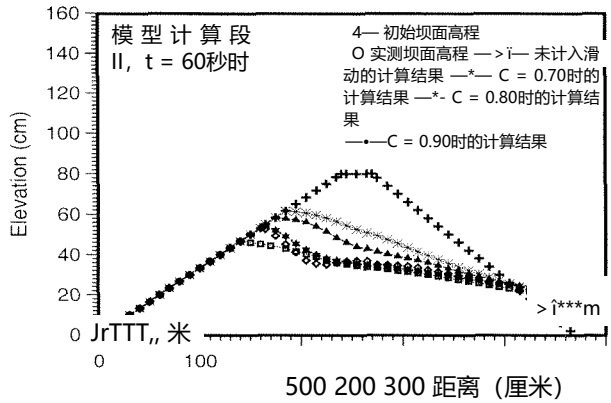


图9 坡面滑动破坏及孔隙水压力系数对计算坝面高程的影响。

数据，且当C = 0.9时，计算所得坝面破坏高程高于实测数据。

模型在实际溃坝事件中的应用

本模型依据辛格（1996）所述实测资料，应用于美国北卡罗来纳州布法罗溪坝因漫顶失效所致溃决出流计算。该坝高32米，长107米，坝顶宽4米，上游坡比为1V:3H，下游坡比为1V:2H。实测最大溃决出流量为1420立方米/秒，出现在漫顶开始后30分钟。当将式（23）中修正斯马特公式内坡度S的指数γ由γ = 0.8降低至0.6（仅适用于S < 0.18情形）时，本模型计算所得溃决出流过程线峰值为实测峰值的1.056倍，峰值出现时间为28分钟。本案例中亦应用了NOAA-BREACH模型（弗莱德，1988）。基于修正迈耶-彼得与穆勒（1948）公式，BREACH模型计算所得溃决出流过程线峰值高达实测峰值的2.8倍，峰值出现时间仅为实测值的0.866倍。基于恩格尔隆德与汉森（1967）全沙输运公式，MIKE-11模型计算所得溃决出流过程线峰值仅为实测峰值的0.167倍，峰值出现时间为实测值的1.167倍。

结论

本研究考虑因漫顶水流引发的坝体侵蚀过程。坝体材料为中等粒径砂土。漫顶断面呈矩形，尾水影响可忽略不计。当发生漫顶时，侵蚀始于坝顶前缘，临界流发生在坝顶前缘下游紧邻断面处。该断面以上为缓流区，基本无侵蚀发生；该临界流断面以下即下游坝面区域为急流区，发生严重侵蚀。

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The MacCormack (1971) explicit finite difference scheme is used to solve the one-dimensional equations of continuity and momentum for unsteady varied flow over steep bed slopes. The comparison between the measured and computed water surface profiles indicates that the hydrodynamic model gives satisfactory results. In the dam erosion computation, the modified Lax scheme with the proposed sediment transport formula gives good results for dam surface elevation compared with the experimental data. The effect of sediment transport formula is very important. Different transport formulas give significantly different erosion results. Good agreement between the computed and measured dam surface profiles was obtained using the modified Smart formula. The inclusion of sliding failure of dam surface slope in the computation significantly improve the model simulation results. The distance-weighting factor α of the modified Lax scheme affects the stability and accuracy of the model. When the dam surface profile is steep, the influence of the bed slope on the computed critical shear stress is large. The opposite condition occurs when the slope of the dam profile is small. The Manning roughness coefficient n and the mean grain size have small effects on the rate of erosion of the dam surface. To accurately compute the dam surface profile due to flow overtopping, analysis of sliding stability is necessary. The empirical pore water pressure coefficient, C_u, is introduced to compute the pore water pressure at a failure slice and C_u is found to be one of the significant parameters. When C_u is increased, the sliding failure increases and the computed dam surface failure drops below the measured data. The opposite condition occurs when C_u is decreased.

The present model computed the breach outflow of the Buffalo Creek dam with a peak equal to 1.056 of the observed value and the time to peak almost the same as the observed value. The NOAA-BREACH model predicted a much higher and earlier peak while the MIKE-11 Dambreak model gave a much lower and later peak.

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采用麦卡马克（1971）显式有限差分格式求解一维非恒定变流连续性方程与动量方程，适用于陡坡床面条件。实测与计算所得水面线对比表明，该水动力学模型结果令人满意。在坝体侵蚀计算中，采用改进型拉克斯格式并结合所提出的泥沙输运公式，所得坝面高程结果与试验数据吻合良好。泥沙输运公式的影响至关重要，不同输运公式给出的侵蚀结果差异显著。采用改进型斯马特公式所得计算坝面高程与实测结果具有良好的一致性。在计算中引入坝面边坡滑动破坏机制，可显著改善模型模拟效果。改进型拉克斯格式中的距离加权因子α影响模型的稳定性与精度。当坝面高程曲线较陡时，床面坡度对计算临界剪应力的影响较大；反之，当坝面高程曲线坡度较小时，该影响则较小。曼宁糙率系数n与平均粒径对坝面侵蚀速率影响较小。为准确计算漫顶水流作用下的坝面高程变化，必须进行滑动稳定性分析。引入经验性孔隙水压力系数C用于计算破坏条块处的孔隙水压力，C被证实为关键参数之一。当C增大时，滑动破坏加剧，计算所得坝面破坏高程低于实测数据；反之，当C减小时，则出现相反情况。

当前模型计算出的布法罗溪坝溃决出流峰值为实测值的1.056倍，峰值出现时间与实测值几乎一致。NOAA-BREACH模型预测的峰值明显更高且提前出现，而MIKE-11溃坝模型则预测出更低且更晚出现的峰值。

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